Telegram AI Assistant Workflow Architecture

Multi-Agent AI System using LangGraph Agents, Tools, and Memory

Soumik Roy

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Contents

1 Introduction

This report explains the complete workflow architecture of a Telegram-based AI Assistant system built using:

- LangGraph AI Agents
- Tool Calling Architecture
- Persistent Memory
- Web Search
- Telegram Automation

The assistant is capable of:

- Conversational interaction
- Real-time web search
- Autonomous tool usage
- Memory-based conversation handling
- Multi-agent orchestration

2 Overall Workflow Architecture

The system follows a modular agent-tool architecture.

2.1 High-Level Data Flow

1. User sends a message through Telegram.
2. Telegram Trigger receives the message.
3. Main AI Agent processes the request.
4. AI Agent decides whether tools are required.
5. Connected tools execute tasks.
6. AI Agent combines results.
7. Final response is sent back to Telegram.

3 Workflow Diagram Placeholder

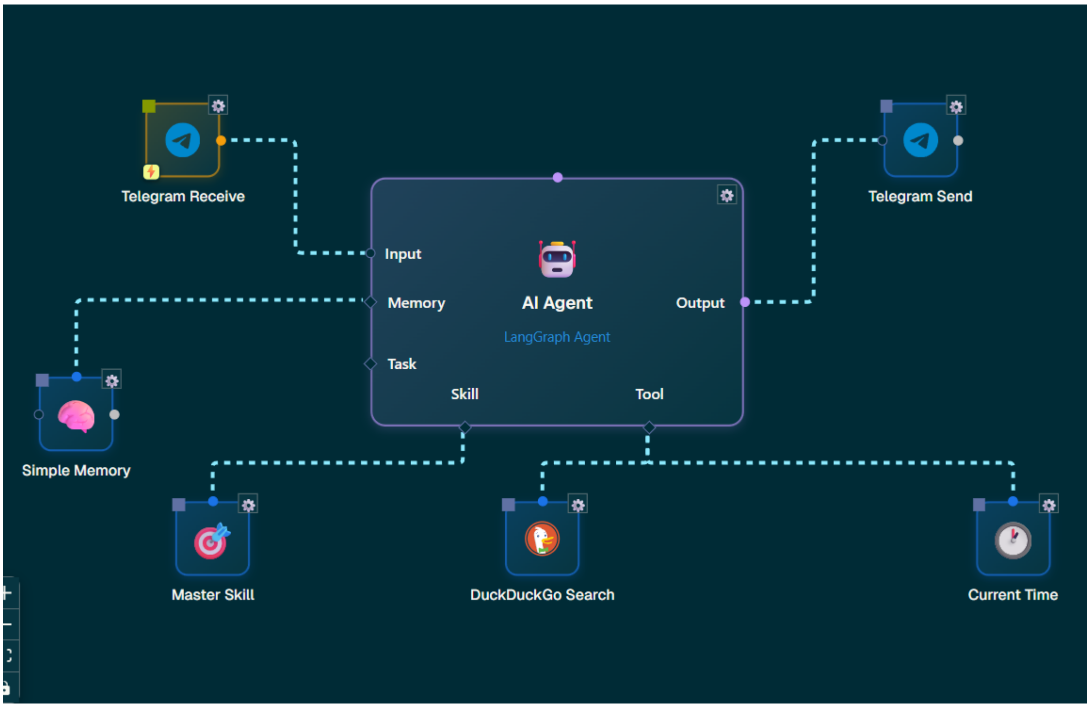

Figure 1: Complete AI Workflow Architecture

4 Core Components

4.1 Telegram Receive Node

The Telegram Receive node acts as the primary trigger for the workflow.

4.1.1 Responsibilities

- Receives incoming Telegram messages
- Extracts metadata:
  - Chat ID
  - Username
  - Message Text
  - Timestamp
- Sends user input to AI Agent

4.1.2 Important Variables

```
{{telegramreceive.text}}
{{telegramreceive.chat_id}}
{{telegramreceive.username}}
```

5 AI Agent Configuration

The AI Agent is the central orchestration engine.

5.1 Purpose

The AI Agent:

- Understands user intent
- Performs reasoning
- Decides when to use tools
- Combines outputs
- Generates final responses

5.2 Model Configuration

- Provider: Gemini
- Model: Gemini Flash Latest
- Temperature: 0.6

5.3 System Prompt

```
You are a helpful Telegram AI assistant.

When needed use available tools whenever needed
to answer questions with current or real-time information.



Reply clearly and concisely.
```

6 AI Agent Configuration Screenshot

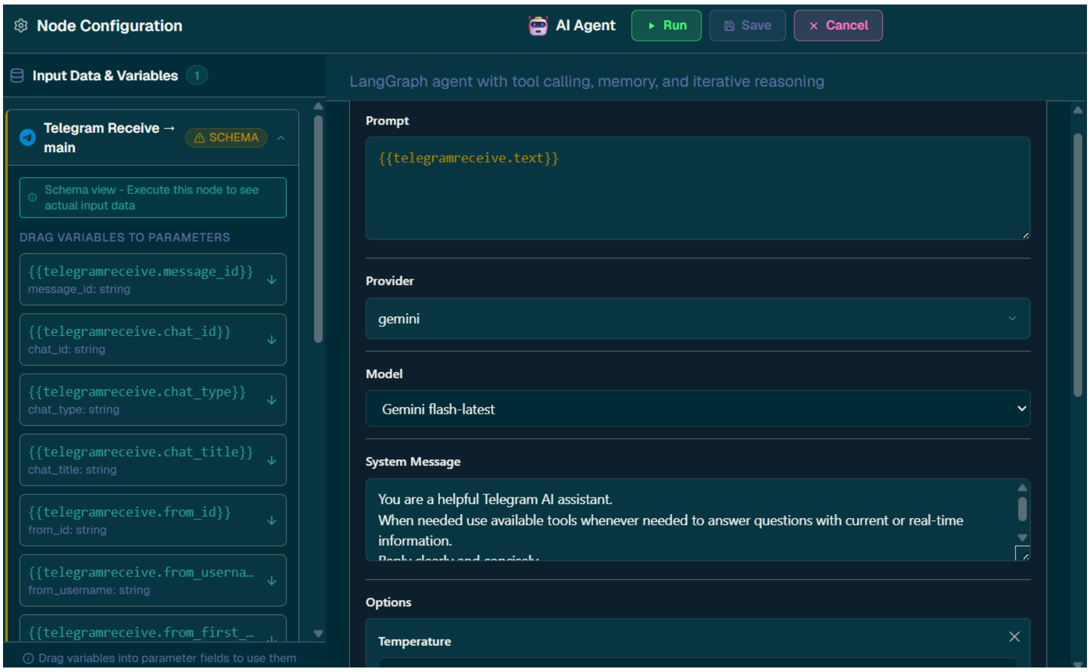

Figure 2: AI Agent Node Configuration

7 Simple Memory Node

The memory node provides short-term contextual memory.

7.1 Purpose

- Maintains conversation continuity
- Stores previous messages
- Enables contextual replies

7.2 Configuration

- Session ID: `{{telegramreceive.chat_id}}`
- Window Size: 100

8 DuckDuckGo Search Tool

This tool provides internet search capability.

8.1 Capabilities

- Real-time web search
- Link retrieval
- Information lookup
- News extraction

8.2 Use Cases

- Song links
- News retrieval
- Trending tweets
- Research tasks

9 Current Time Tool

Provides real-time date and time context.

9.1 Use Cases

- Time-aware responses
- Scheduling
- Date calculations

10 Master Skill Node

The Master Skill node enables task decomposition.

10.1 Purpose

- Delegates tasks
- Handles orchestration
- Coordinates tool usage

11 Telegram Send Node

The final output node.

11.1 Responsibilities

- Sends final responses to Telegram
- Formats messages
- Delivers outputs

11.2 Message Variable

```
{{aiagent.response}}
```

12 Pseudo Code

12.1 Overall Workflow Pseudocode

```
START

Receive Telegram Message

Store message in memory

Send message to AI Agent

AI Agent:
    Analyze user intent

    IF web information needed:
        Use DuckDuckGo Search


    Combine tool outputs

Generate final response

Send response via Telegram

END
```

13 Observed Outputs

13.1 Music Link Retrieval

The AI successfully:

- Understood the requested song

- Searched the internet
- Retrieved YouTube links
- Returned structured output

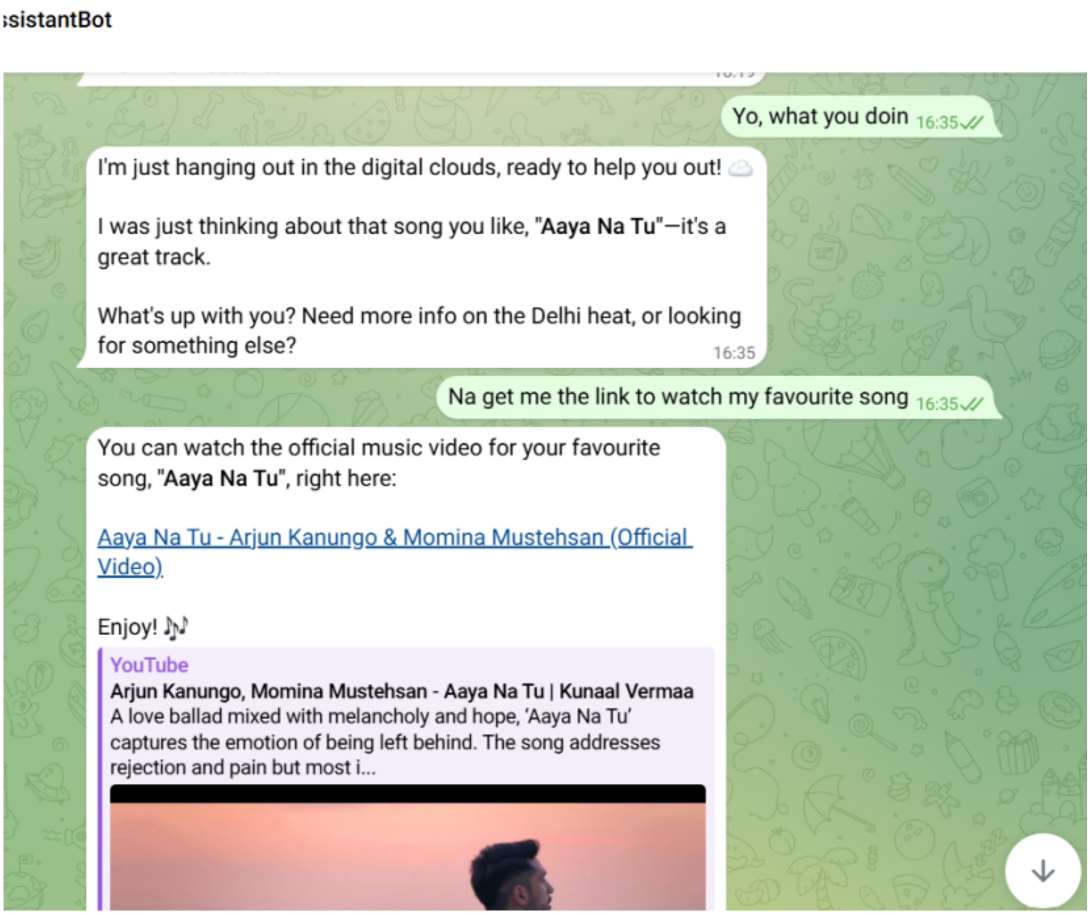

Figure 3: Song Search Example

14 Viral Tweet Retrieval

The AI successfully:

- Performed web search
- Retrieved trending tweets
- Summarized findings
- Provided direct links

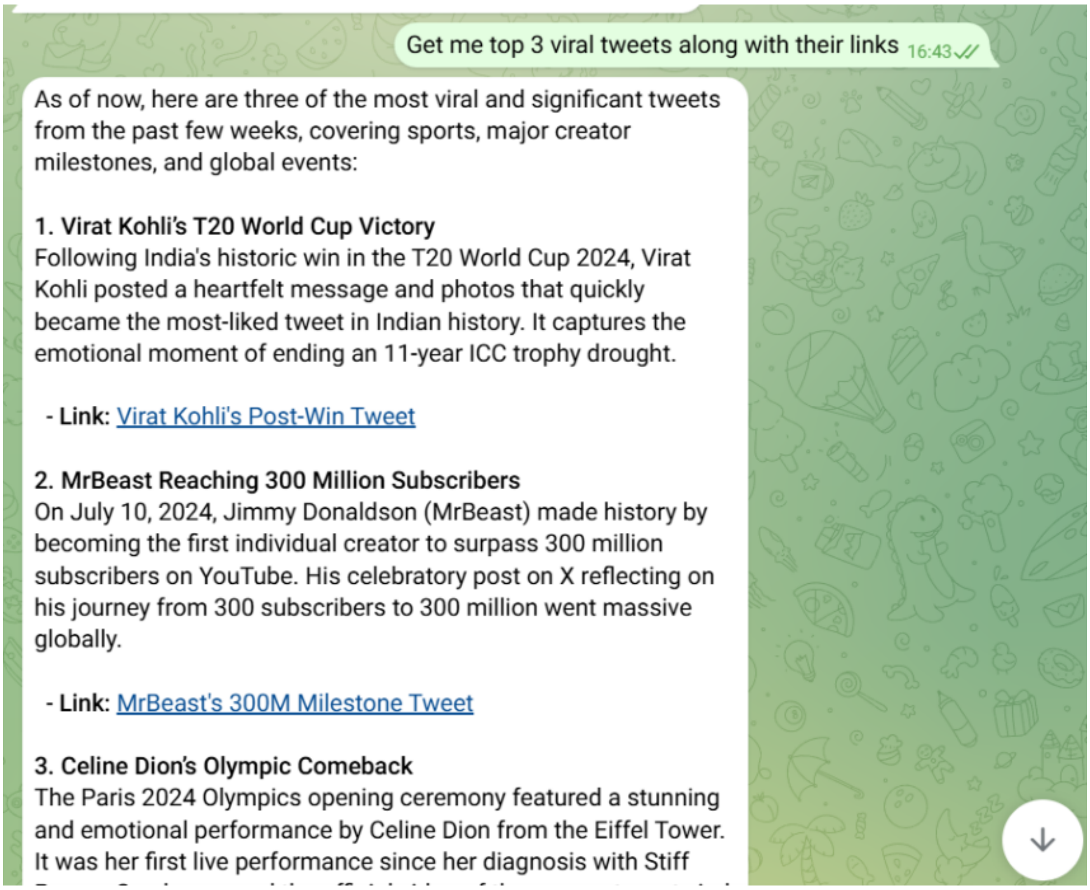

Figure 4: Trending Tweet Retrieval Example